

Module 7

Explaining Application Services

Learning Objectives

- Explain the importance of time synchronization and the role of NTP.
- Explain the use of web, file/print, and database services.
- Explain the use of email and voice services.
- Explain how high availability services are provisioned using redundancy and load balancing.

Lesson 7.1

Application Security and Time Synchronization

Transport Layer Security (TLS)



Encryption

- Encrypts the data

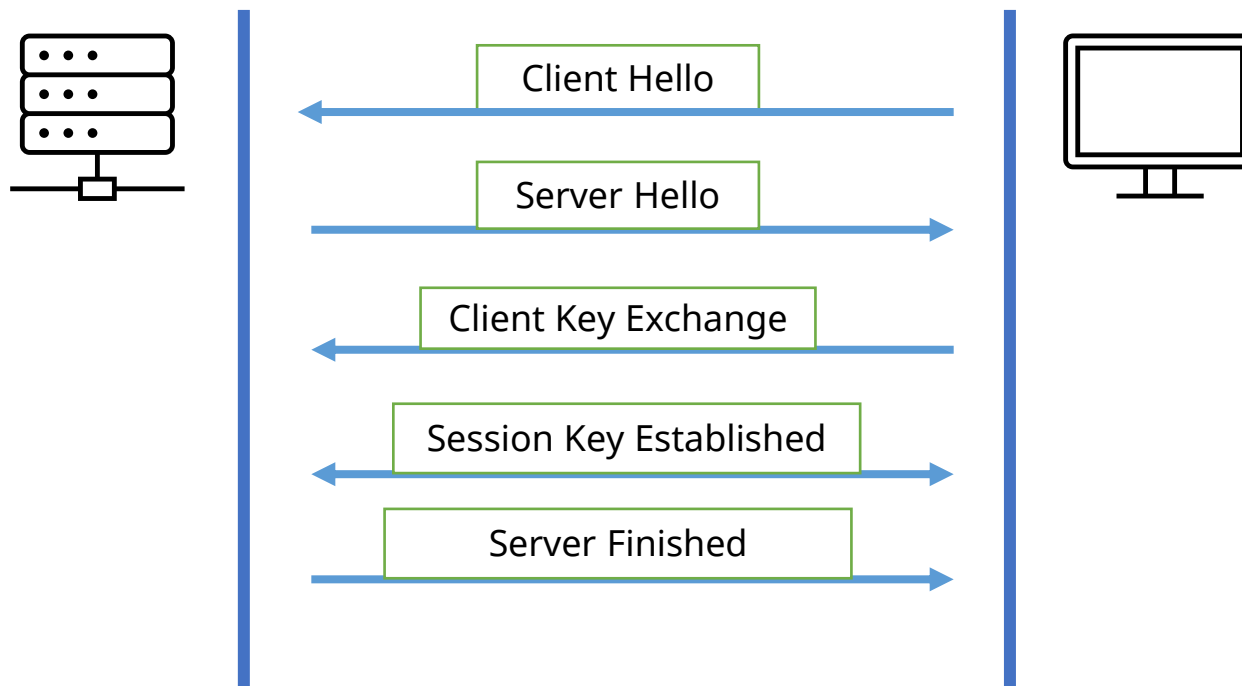
Authentication

- Confirm server identity with certificates

Integrity

- Ensures data is unchanged

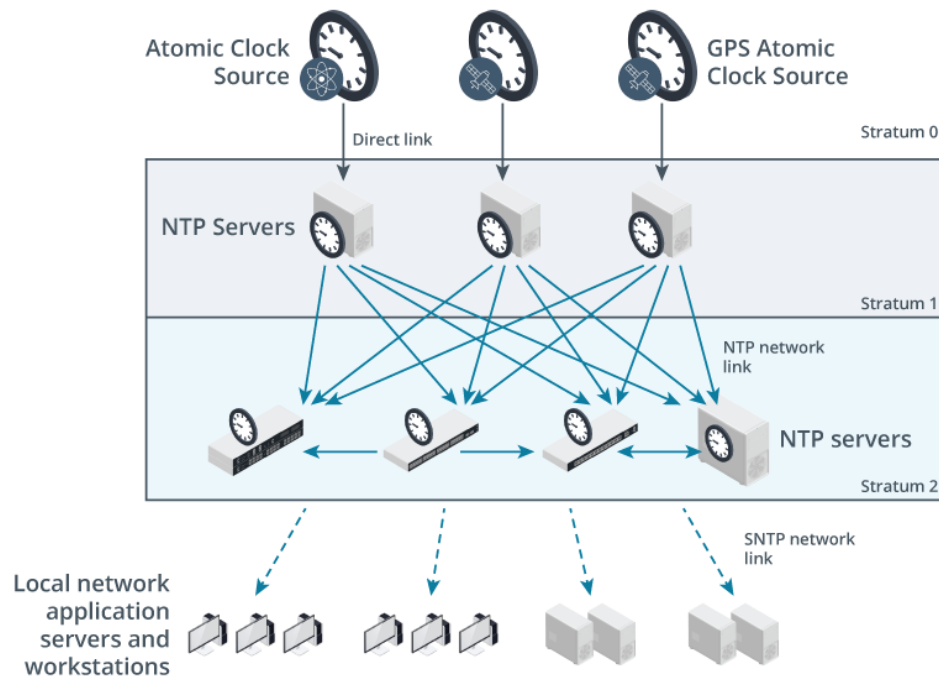
TLS Handshake



Network Time Protocol (NTP)

Purpose	Operation	Hierarchy	Synchronization	Algorithms
• Synchronizes system clocks to Coordinated Universal Time (UTC)	• Operates over networks using the UDP protocol on port 123	• Organizes devices in a hierarchical manner, indicating distance from the clock	• Synchronizes devices by sending timestamped messages to/from network peers	• Uses complex algorithms to account for network delay and jitter

NTP Diagram



Precision Time Protocol (PTP)

Applications

- Industrial processes
- 5G cellular data
- Medical devices
- Financial services
- Broadcasting

Clock types

- Grandmaster clock
- Boundary clock
- Ordinary clock

Roles

- TimeTransmitter
- TimeReceiver
- Grandmaster
- Boundary

Activity: Fill in the Blank

- TLS _____ the data to hide it from anyone who might intercept it.
- _____ is used to synchronize clocks to Coordinated Universal Time.
- The _____ clock is the main time source in the PTP domain.



Activity: Fill in the Blank

- TLS encrypts the data to hide it from anyone who might intercept it.
- NTP is used to synchronize clocks to Coordinated Universal Time.
- The Grandmaster clock is the main time source in the PTP domain.



Lesson 7.2

Web, File/Print, and Database Services

Hyper Text Transfer Protocol (HTTP)

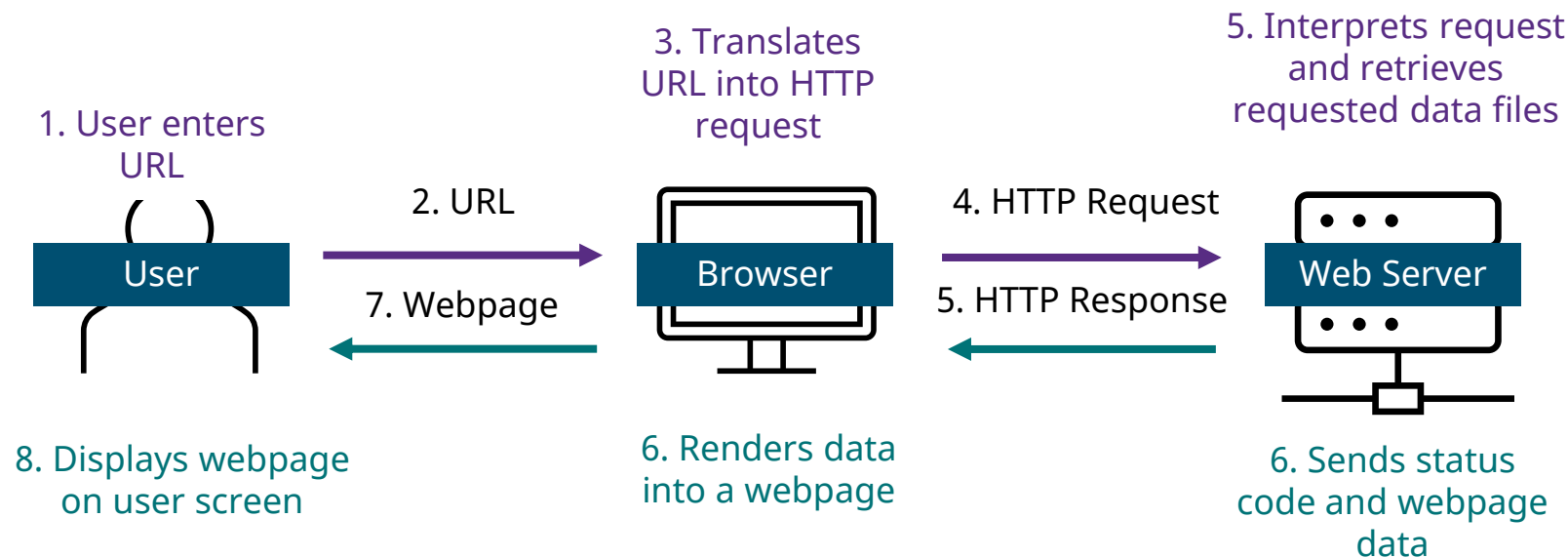
Purpose

- Protocol for fetching resources (HTML documents, images, and videos) from the web

Foundation

- Basis of any data exchange on the web

Hyper Text Transfer Protocol (HTTP)



HTTP Secure (HTTPS)

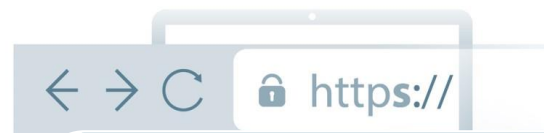


HTTP

- Foundational protocol for the Internet
- Does not encrypt data
- Uses port 80
- URL begins with http://



secure



HTTPS

- Secure version of HTTP
- Uses TLS (SSL) encryption
- Uses port 443
- URL begins with https://

File Transfer Protocol (FTP)

FTP Server Authentication

- Require users to provide credentials
- Some allow anonymous access

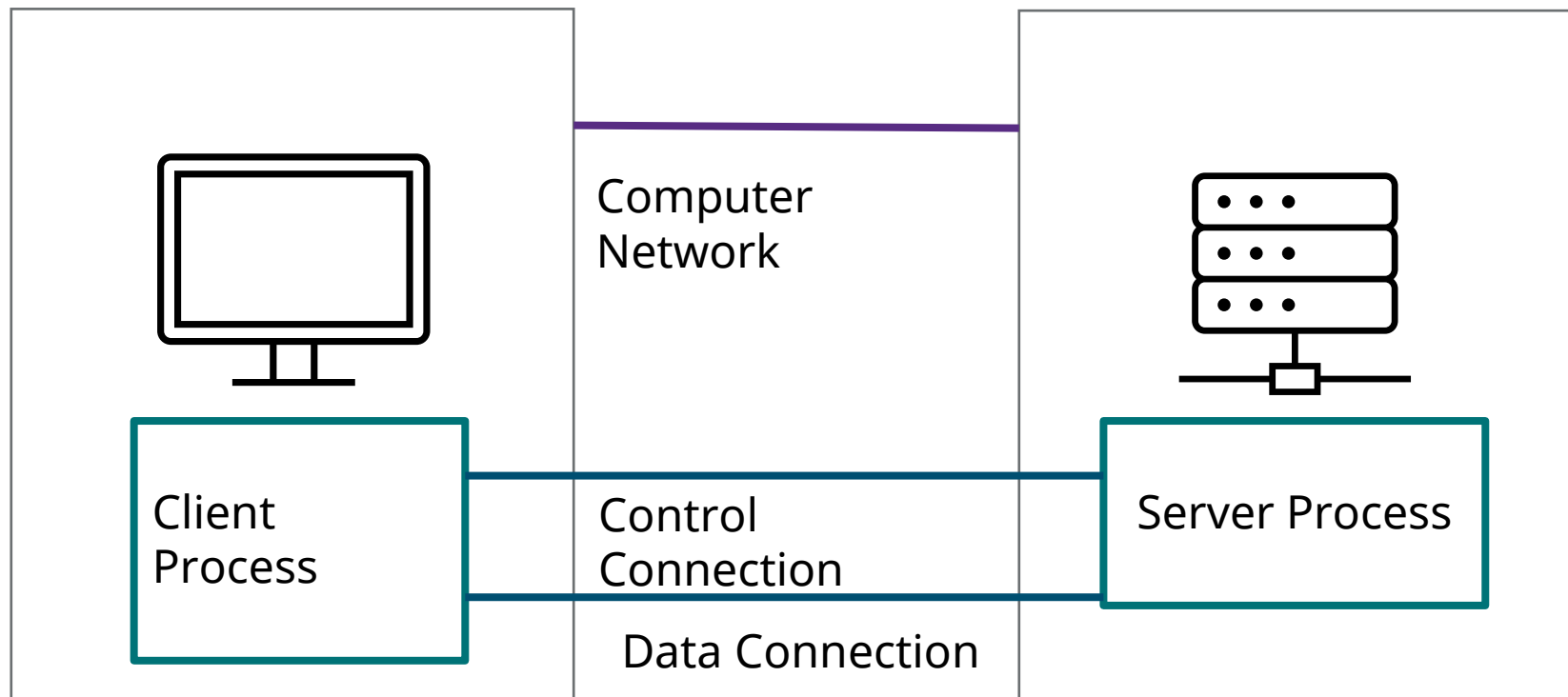
Channels

- Command channel for instructions and response
- Data channel for transferring files

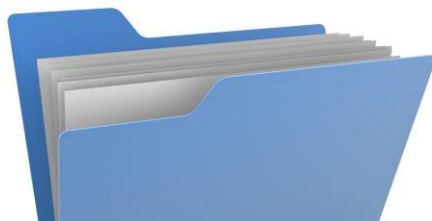
Transfer Modes (Active and Passive)

- Active: server connects back to the client to transfer data
- Passive: client initiates command/data connections to avoid firewall issues

FTP Diagram



Secure File Transfer Protocol (SFTP)



FTP

- Unencrypted data
- Uses two channels
- Uses ports 20 and 21



SFTP

- Encrypted data
- Uses one channel
- Uses port 22

Server Message Block Protocol (SMB)

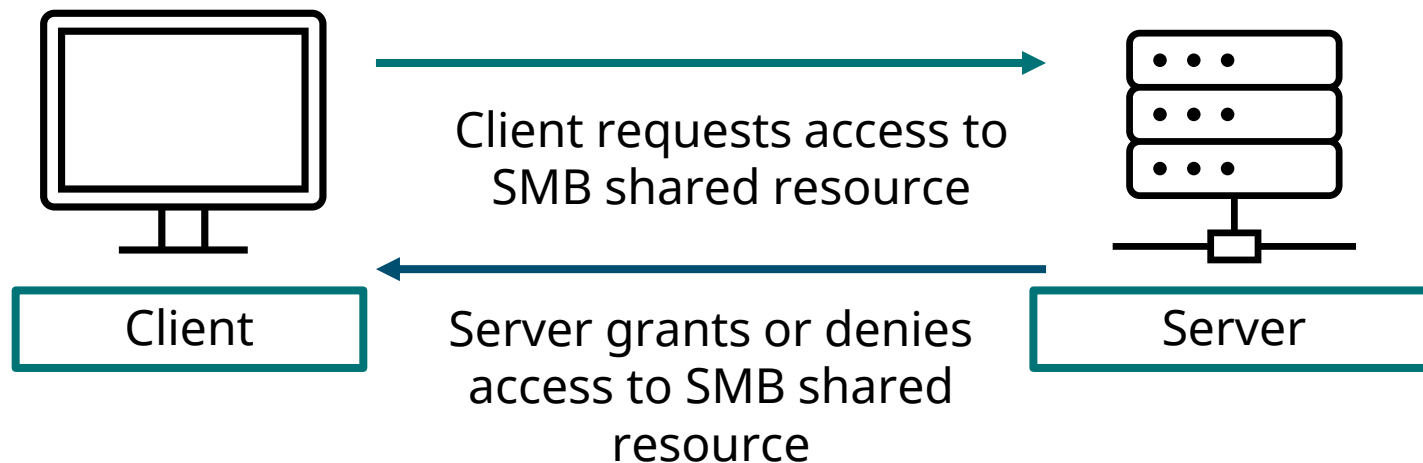
SMB Purpose

- Allows computers in the same network to share resources

SMB Usage

- Windows
- Linux using Samba

SMB Diagram



Network Attached Storage (NAS)

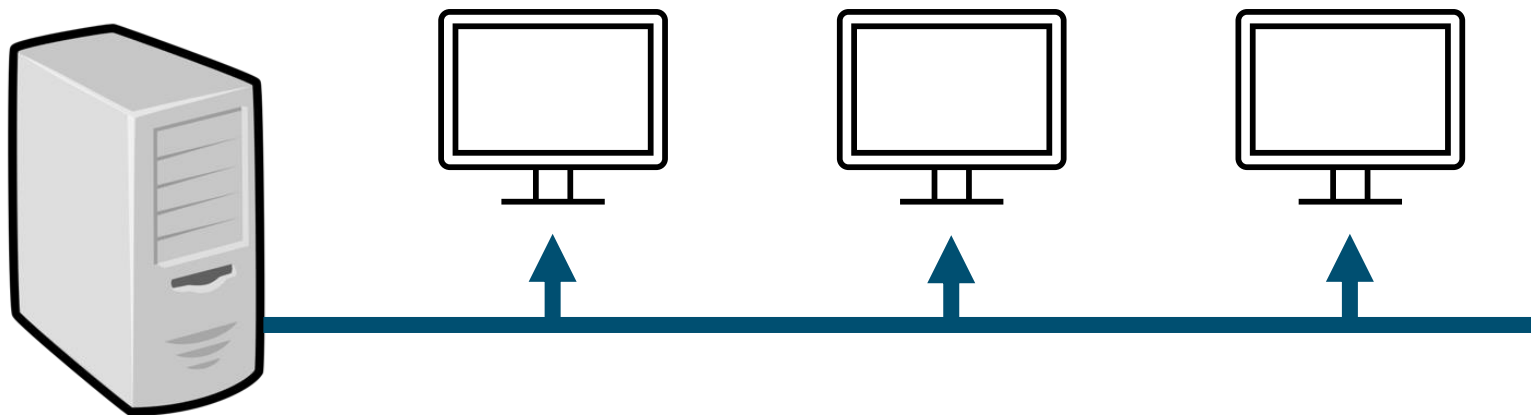
NAS Purpose

- Provides remote file access
- Concurrent resource access for multiple devices

NAS Components

- Processor, memory, and hard drives
- Bespoke OS, usually Linux

NAS Diagram



RDBMS Database Service

Relational Database Management Systems

- Structured storage (tables store data)
- SQL queries to retrieve/update data
- Connected tables link data via unique IDs
- Encryption via TLS

NoSQL Database Service

NoSQL Databases

- Stores data in different formats (key-value pairs/wide columns)
- Simple Access through APIs over HTTPS
- Scalable Growth (Horizontal scaling)
- Supports wide range of data models

RDBMS vs NoSQL

RDBMS

Requires predefined schema

Vertically scalable

Handles complex queries

NoSQL

Schema-less or dynamic schemas

Horizontally scalable

Simplicity/speed for specific types of queries

Activity: Matching

HTTP

Protocol for web communication

FTP

Storage accessible over a network

SMB

Network protocol for file transfer

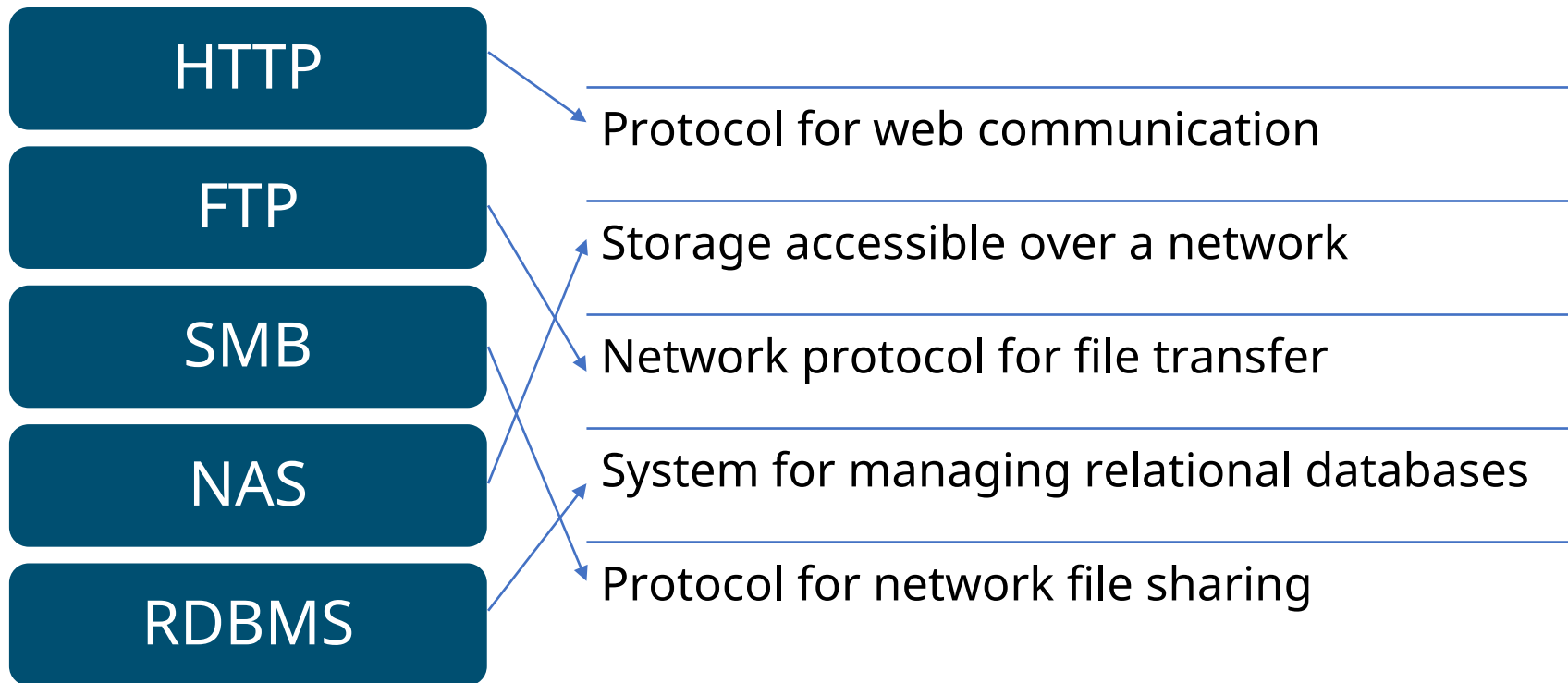
NAS

System for managing relational databases

RDBMS

Protocol for network file sharing

Activity: Matching



Lesson 7.3

Email and Voice Services

Simple Mail Transfer Protocol (SMTP)

Purpose

- Used to send emails from one server to another

Process

- Routes emails from sender's email client to recipient's email server

Security

- Combinable with TLS to encrypt emails during transmission

Ports

- Port 25, port 465 with implicit TLS, port 587 with explicit TLS

Internet Message Access Protocol (IMAP)

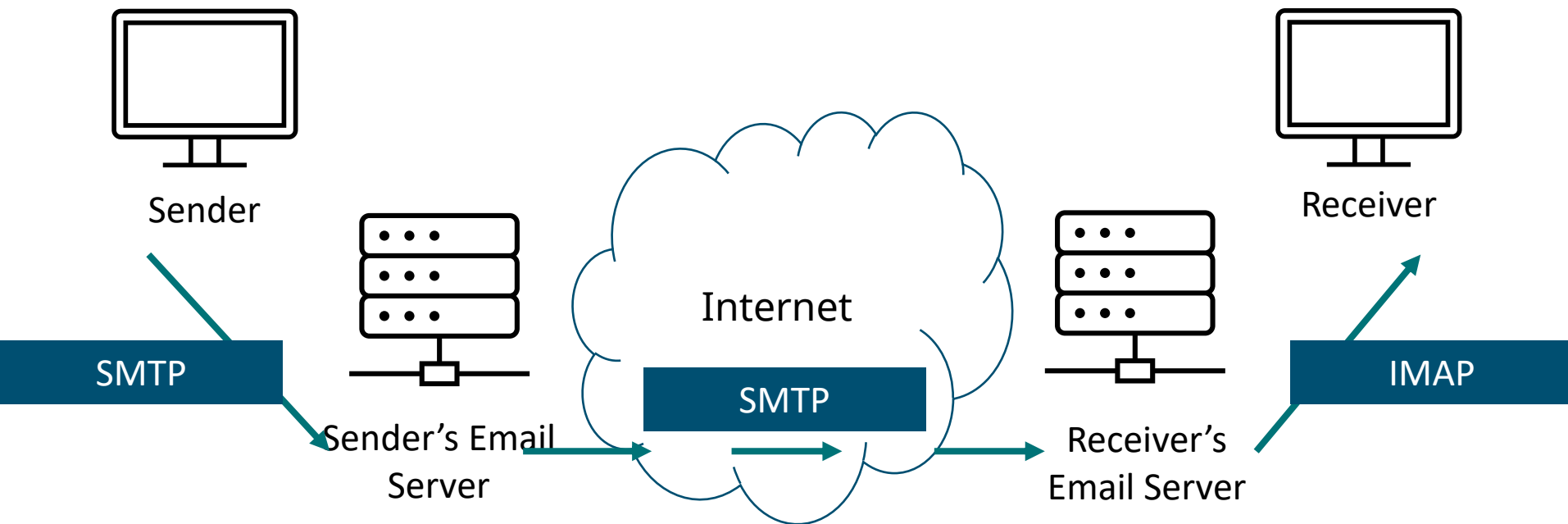
Purpose

- Access and manage email remotely
- Stores emails on server

Function

- Connecting to the mail server
- Synchronization
- Reading emails
- Managing emails

SMTP and IMAP Diagram



Voice and Video Services

VOIP

- Transmits voice calls over IP networks
- Supports voice, video, and multimedia sessions
- Supports various devices

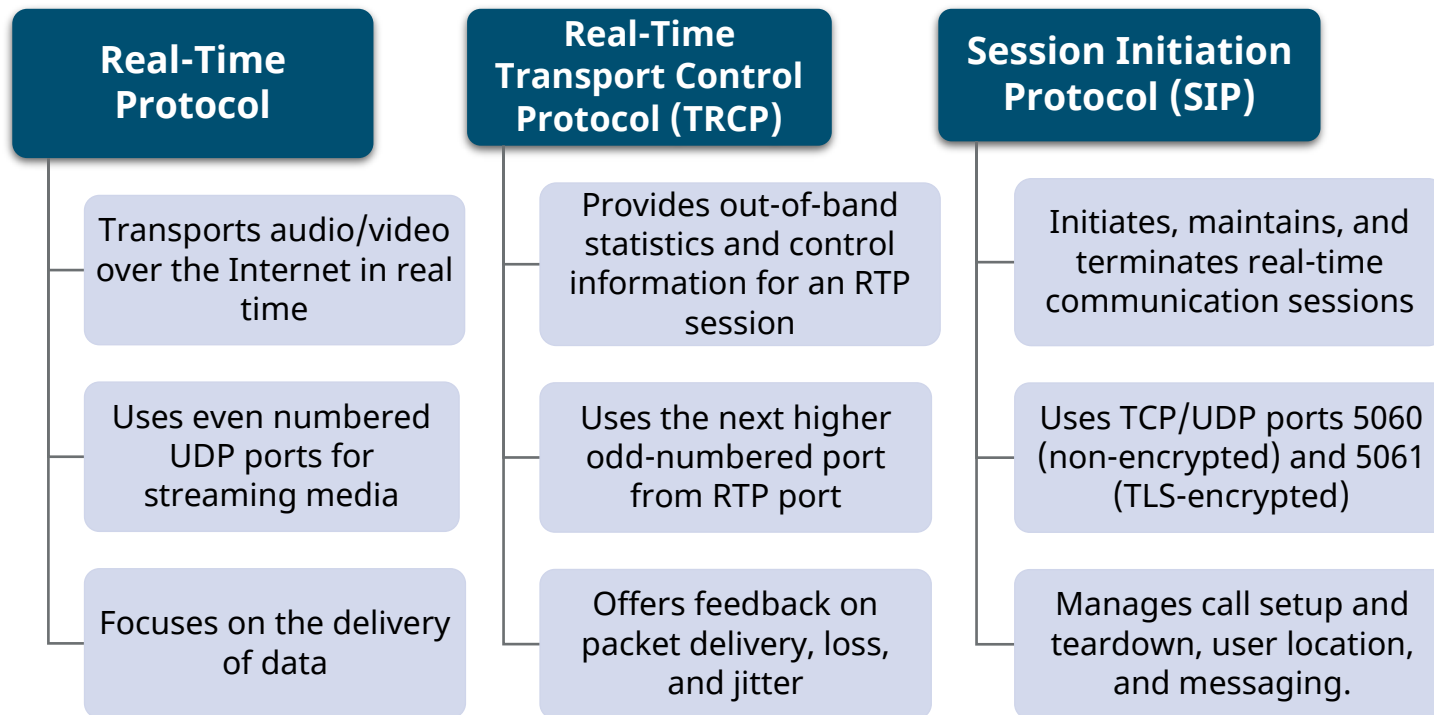
PBX

- Private Branch Exchange
- Connects internal phone lines within an organization
- Manages incoming/outgoing calls, voicemail, and call forwarding

VoIP-Enabled PBX

- Combines PBX functions with VoIP
- Operates over LAN or WAN
- Reduces reliance on physical infrastructure

VoIP Protocols



Quiz: Which protocol allows you to access and manage emails remotely?

- a) RDP
- b) RTP
- c) IMAP
- d) SMTP

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Quiz: Which protocol is used to send emails from one server to another?

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Lesson 7.4

Disaster Recovery and High Availability

Disaster Recovery Concepts

Disaster Recovery Plan (BRP)

Guidelines and procedures

Preparation and prevention

Tabletop Exercises

Simulation

Constrictive learning

Validation Tests

Functional testing

Scenario testing

Full-scale testing

Business Continuity Plan (BCP)

BCP team

Business impact analysis (BIA)

Crisis communication plan

Disaster Recovery Metrics

High availability (HA)

- Ensures consistent system uptime

Maximum tolerable downtime (MTD)

- Longest disruption without severe impact

Recovery time objective (RTO)

- Time to restore services after disruption

Work recovery time (WRT)

- Time to confirm system/data integrity post-recovery

Recovery point objective (RPO)

- Maximum acceptable loss after an event

Disaster Recovery Sites



Cold Site



Warm Site



Hot Site

Fault Tolerance and Redundancy

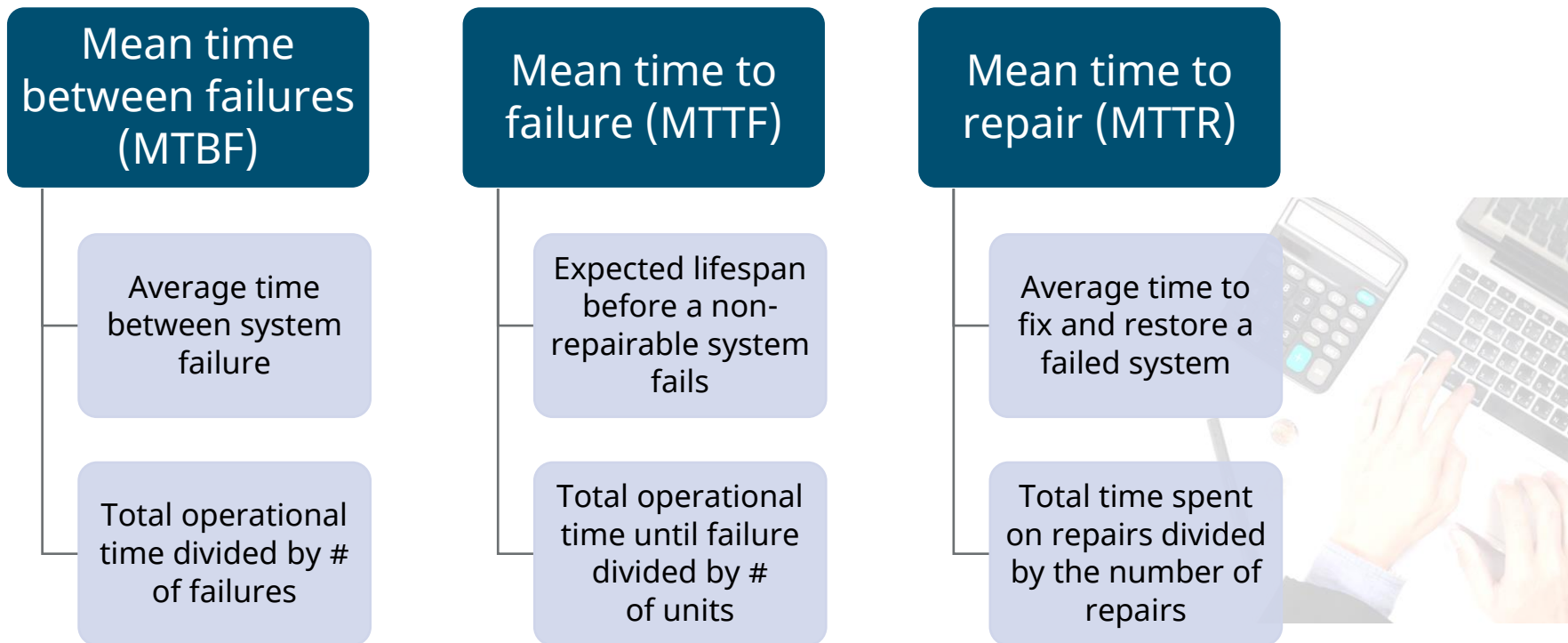
Fault Tolerance

- Maintains operation despite component failure
- Operates at reduced capacity instead of shutting down
- Uses replication, error checking, and failover

Redundancy

- Duplicates key components for higher reliability
- Offers backup paths/capacity for failed components
- Prevents system collapse from any single failure

Key Performance Indicators (KPIs)



Load Balancers

Purpose

Distributes network traffic across multiple servers

Improves application performance and reliability

Functionality

Acts as a “traffic cop” to route client requests

Ensures no single server is overworked

Benefits

Reduces downtime and increases scalability

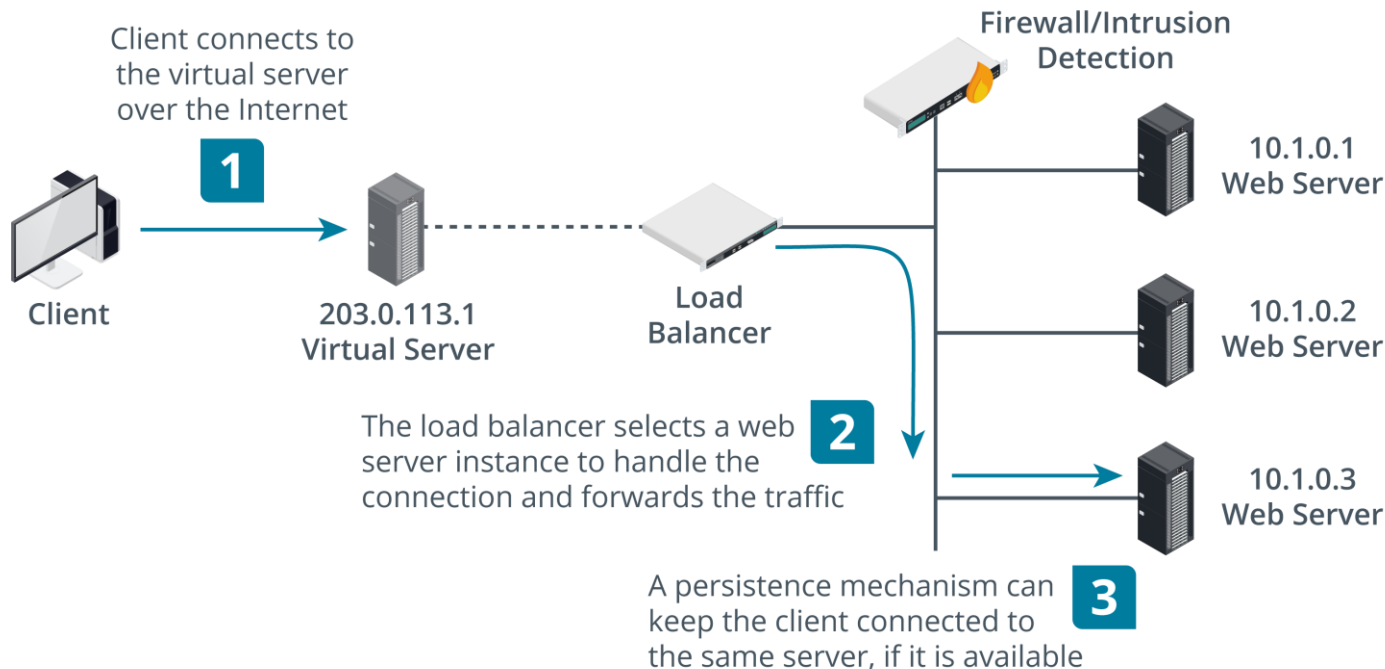
Provides redundancy/efficiency in handling requests

Types

Can be hardware-based or software-based

Uses various algorithms like Round Robin, Least Connections, etc.

Load Balancer Diagram



High Availability Clusters

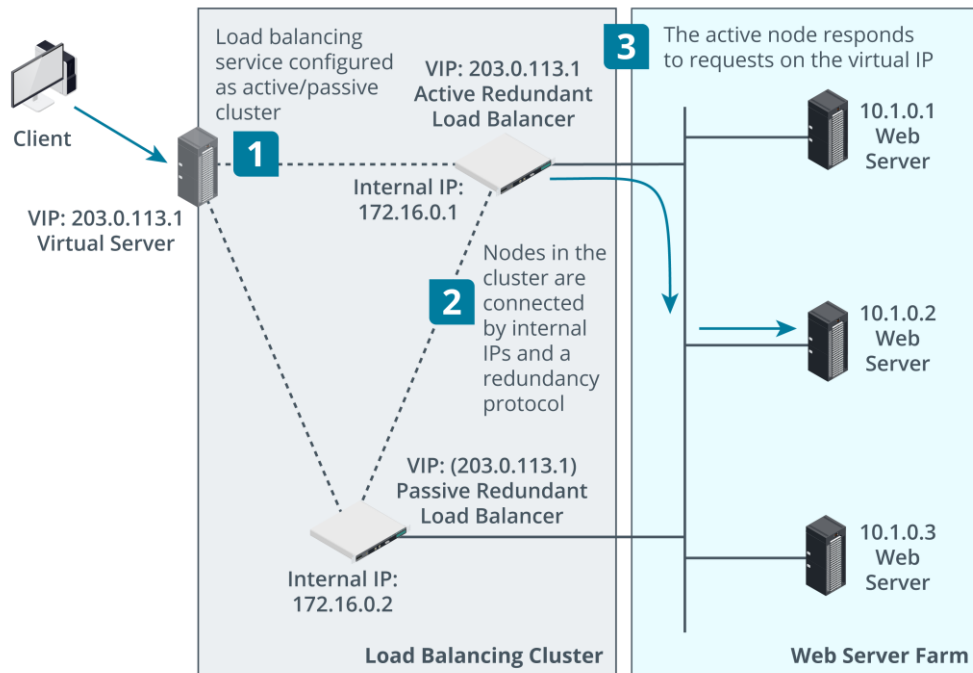
High Availability Clusters Definition

- Systems that ensure continuous operation and minimize downtime via redundant components and failover mechanisms

High Availability Clusters Function

- Ensure continuous service/minimal downtime for critical applications
- Automatically manage failovers to maintain operations
- Monitor system health and provide redundancy
- Ideal where uptime is crucial (e.g., databases, business services)

High Availability Clusters Diagram



First Hop Redundancy

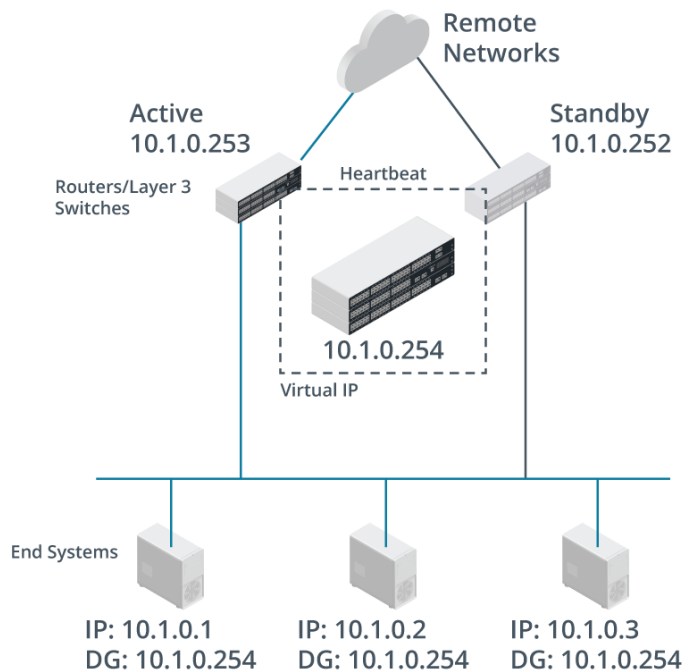
First Hop Redundancy Definition

- Ensures uninterrupted network service by automatically switching to a backup gateway if the primary one fails

Functions

- Ensures uninterrupted gateway availability
- Multiple routers prevent single points of failure
- Quick failover with protocols like HSRP/VRRP
- Maintains network access and reliability

First Hop Redundancy Diagram



Activity: Hot, Warm, or Cold?

A financial services company that processes high-volume transactions cannot afford more than a few minutes of downtime. Any downtime can result in significant financial loss and erosion of customer confidence.

Which type of recovery site would be best for their situation?



Cold Site



Warm Site



Hot Site

Summary

- **HTTPS everywhere:** Implement HTTPS (TLS/443) on web servers for secure communication
- **Secure file transfer:** Utilize secure protocols like SFTP/FTPS (SSL/TLS) for file transfers
- **Limit unencrypted sharing:** Restrict unencrypted services (SMB) to trusted internal networks
- **Plan for disruptions:** Develop DR/HA plans with dedicated resources for business continuity